

EMISSIONS

- RFF-SP probabilistic ensemble
- 10 000 socioeconomic + emissions trajectories
- Rennert et al. 2022

CLIMATE

- FaIR v2.2.4, 841 v1.4.1-calibrated config*
- Stochastic internal variability
 - GMST(t), OHC(t) trajectories
 - Smith et al. 2024

SEA LEVEL

- MimiBRICK 10k posterior subsample
- Importance weighting (Wong 2026)
- 1850–2300 horizon

DAMAGE-FUNCTION TRANSLATION (FrEDI)

Sweet et al. (2022) NCA5 scenarios as interpolation nodes (0, 30, 50, 100, 150, 200, 250, 300 cm by 2100)
11-year rolling average to smooth capital-investment lumps • Linear interpolation between bracketing SLR-cm scenarios per year

Coastal Properties

Neumann (2021) + Lorie (2020)
NCPM benefit-cost decisions:
armor / elevate / nourish / abandon
Variant: Reactive

HTF Transportation

Fant (2021)
Passenger + freight delay costs;
elevated-road & seawall protection
Variant: Reasonably Anticipated

HTF Elder Mortality

Sheahan et al. (2025)
Mueller (2024) flood-depth → mortality
relationship; ICLUS pop × BenMAP
VSL = 7.9M(1990)

→ **STATE-LEVEL DAMAGES PER (YEAR, SCENARIO, ADAPTATION VARIANT)** ←

METHODS: Each RFF-SP emissions trajectory drives a stochastic FaIR climate run; GMST(t) and OHC(t) feed MimiBRICK to give a probabilistic global SLR trajectory. BRICK draws are importance-weighted against observed GMSL (Wong 2026 AR(1) likelihood). Per-year global SLR is matched to bracketing Sweet (2022) NCA5 nodes; FrEDI interpolates linearly between bracket-specific damage functions and applies an 11-yr rolling average to smooth lumpy capital investments. Output: per-(year, sector, state, scenario, adaptation variant) damages in 2015 USD.